WT-90 Wind Turbine Maintenance Manual

1 Safety

No one may climb the tower when the wind speed at hub height exceeds 15 metres per second. Two technicians are always present, and each carries a fall arrest harness and a radio.

2 Intervals

A visual inspection takes place every six months and a full inspection every year. The gearbox oil is changed every three years, or earlier if the oil analysis shows particles above the limit.

3 Annual inspection

3.1 Procedure

The annual inspection follows the steps below, in this order:

1. Stop the rotor with the service brake and engage the rotor lock.
2. Switch the controller to maintenance mode with the command shown in section 3.2.
3. Check every blade bolt with a calibrated torque wrench, at the torque given in Table 2.
4. Record the readings in the maintenance log and release the rotor lock.

3.2 Controller commands

The controller is operated from the console in the nacelle:

```
wtctl mode maintenance --confirm
wtctl status
mode: maintenance
rotor_lock: engaged
```

4 Reference values

Table 2 lists the tightening torques of the main bolted joints.

Joint	Torque (N m)
Blade bolts	1,450
Tower flange	2,900
Yaw bearing	980

Table 2. Tightening torques.